

Mortgage Comparison Analysis

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Financial Mathematics

Universe

- Loan amount: \$320,000
- Options: \{30-Year Fixed 6.5%, 15-Year Fixed 5.8%, Refinance 30yr->25yr at 5.2%\}

Variables

Symbol	Meaning	Type / Range
L	loan principal = \$320,000	$\mathbb{R}_{>0}$
r	monthly interest rate = APR/12	$\mathbb{R}_{>0}$
n	total number of monthly payments	$\mathbb{Z}^+$
PMT	monthly payment amount	$\mathbb{R}_{>0}$

Structure

A homebuyer is purchasing a \$400,000 home with an \$80,000 down payment (\$320,000 loan). Compare three mortgage options:

1. **30-year fixed** at 6.5% APR
2. **15-year fixed** at 5.8% APR
3. **Refinance scenario**: Start with the 30-year at 6.5%, then after 5 years refinance the remaining balance into a new 25-year loan at 5.2% APR with \$6,000 in closing costs

Which option minimizes total cost? When does refinancing break even against staying with the original 30-year loan?

Real-World Mapping

Real-World Concept	Mathematical Object
loan balance	L — principal
annual percentage rate	r — monthly rate = APR/12
loan term	n — number of monthly payments
monthly mortgage payment	PMT — annuity payment

Constraints

(1)

$$PMT = L \cdot \frac{r(1+r)^n}{(1+r)^n - 1}$$

annuity payment formula

(2)

$$B_k = B_{k-1}(1+r) - PMT$$

amortization recurrence

(3) $B_n = 0$

loan fully repaid at maturity

(4)

$$\text{Total Interest} = n \cdot PMT - L$$

total interest = payments minus principal

Objective

$$\min_{\text{option}} \sum_{k=1}^n PMT_k + \text{closing costs}$$

Approach

Suggested approach Annuity PMT + Amortization Schedule + Break-Even Search

Complexity class:

Available tools: Annuity PMT + Amortization Schedule + Break-Even Search

2. Solution

Answer:	cheapest: 15-Year Fixed 5.8%; refinance break-even at month 26
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	Annuity PMT + Amortization Schedule + Break-Even Search
Time:	0.0023s
Certificate:	Amortization verified: final balance = \$0.00 for all options.

Solution Details

Home price: \$400,000
Down payment: \$80,000
Loan amount: $L = \$320,000$

Mortgage Options Comparison:

30-Year Fixed 6.5%:
Monthly payment: $PMT = \$2,022.62$
Total interest: \$408,142.38
Total paid: \$728,143.20

15-Year Fixed 5.8%:
Monthly payment: $PMT = \$2,665.89$
Total interest: \$159,859.70
Total paid: \$479,860.20

Refinance 30yr->25yr at 5.2%:
Monthly payment: $PMT = \$1,786.25$
Total interest: \$337,232.57
Total paid: \$663,232.20

Cheapest option: 15-Year Fixed 5.8%
Refinance break-even: month 26 (2.2 years)
Refinance net savings: \$64,911

Verification

No independent verification checks recorded.

3. Interpretation

The Question

A homebuyer is purchasing a \$400,000 home with an \$80,000 down payment (\$320,000 loan).

The Answer

Cheapest option: 15-Year Fixed 5.8%; refinancing saves \$64,911; break-even at month 26.

What This Means

- Monthly payments for each option
 - Full amortization schedule summaries (total interest, total paid)
 - Refinance break-even month (when cumulative savings exceed closing costs)
 - Net savings from refinancing over the remaining loan life
 - Independent verification that amortization schedules sum correctly
1. PMT calculation: Compute monthly payment using the annuity formula $P = L * r(1+r)^n / ((1+r)^n - 1)$
 2. Amortization schedule: For each month, split payment into principal and interest, track remaining balance
 3. Refinance analysis: After 60 months on the 30-year loan, compute remaining balance, then start a new 25-year schedule at the lower rate plus closing costs
 4. Break-even: Find the month where cumulative payment savings from refinancing exceed the closing costs
 5. Verify: Independently confirm that principal payments sum to the loan amount and that final balances are zero

Recommendations

1. Compare total interest paid, not just monthly payments, when evaluating mortgage options.

Limitations

- Assumes fixed interest rates; adjustable-rate mortgages introduce variability.

- Does not
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